

PRANEY GOYAL

praneygoyal@gmail.com | praneyg.github.io | github.com/praneyg

EDUCATION

The Pennsylvania State University

May 2025

Master of Science in Computer Science

GPA: 3.19/4.0

Bachelor of Science (Honors) in Computer Science

GPA: 3.59/4.0

Minor in Mathematics

RESEARCH EXPERIENCE

Research Assistant

Jul 2026 – Present

University College London

Remote

Advisor: Mario Giulianelli | `uncertain` library (ongoing)

- Designing `uncertain`, a Python library for sampling-based uncertainty estimation, unifying surprisal, entropy, and information value within a configurable generalized-surprisal framework.
- Specifying modular APIs for samplers, representation encoders, distances, and statistics, with reusable sampled alternatives intended to support parameter sweeps without repeated model generation.
- Designing reproducibility and estimation interfaces that record sampling provenance, deterministic seeds, standard errors, and truncation diagnostics alongside each uncertainty estimate.

Research Assistant

Jul 2026 – Present

Indian Institute of Technology Roorkee

Remote

Advisor: Raksha Sharma | Appointment through Dec 2026 (expected)

- Implemented experiments and research code for zero-shot fake-news detection across LLaMA-3-8B, Qwen2.5-7B, and Mistral-7B on PolitiFact, GossipCop, and LUN.
- Analyzed layer-wise predictions with the logit lens to localize confidence failures; adversarial publisher-style rewrites increased expected calibration error by 0.23–0.35 across nine model–dataset combinations.
- Evaluated class-conditional accuracy and AUROC to distinguish threshold shifts from ranking failures; tested label-free threshold correction using the benchmarks’ known class balance.
- Co-authored three manuscripts on calibration, robustness evaluation, and mechanisms of style-induced failures; under review at ACML, ACL ARR, and WSDM.

DL/AI Researcher

Jul 2025 – Jun 2026

Brown University

Remote

- Investigated how spurious tokens manipulate predictions in LLMs adapted with low-rank adaptation (LoRA), studying robustness across model architectures, datasets, and corruption conditions.
- Built reproducible evaluation and diagnostic tools to measure adversarial correlations and assess preprocessing, paraphrasing, and grammar-based filtering as mitigation strategies.
- Co-authored workshop papers at NeurIPS 2025 and ICLR 2026 on spurious-token vulnerabilities and safer LLM fine-tuning.

Machine Learning Research Assistant

Aug 2023 – May 2025

RAISE Lab, Pennsylvania State University

University Park, PA

- Developed a GPT-4o-powered counterfactual explanation framework for NLP models, improving non-technical stakeholder trust scores by 35%.
- Improved model accuracy by 10.4% through counterfactual data augmentation evaluated across six machine learning models and three datasets.
- Created an interpretability evaluation methodology combining automated metrics with LLM-based validation.

PUBLICATIONS AND MANUSCRIPTS

Under review

A Layer-wise Study of How Style Rewriting Breaks the Calibration of Zero-Shot LLM Fake-News Detectors

Prabhjot Singh, **Praney Goyal**, Raksha Sharma.
Under review, Asian Conference on Machine Learning (ACML), 2026.

Broken Model or Broken Metric? Rethinking Robustness Evaluation in Fake Detection

Prabhjot Singh*, **Praney Goyal***, Raksha Sharma.
Under review, ACL Rolling Review (ARR), August 2026 cycle.

Digging Deeper for Truth: Explaining Why Style Rewriting Defeats Zero-Shot LLM Fake-News Detection

Prabhjot Singh*, **Praney Goyal***, Raksha Sharma.
Under review, 20th ACM International Conference on Web Search and Data Mining (WSDM).

Workshop publications

LORA users beware: A few spurious tokens can manipulate your finetuned model

M. M. Salles*, **P. Goyal*** et al.
ICLR 2026 Workshop on Principled Design for Trustworthy AI.

Paraphrasing Away Malicious Tokens: Improving LLM Finetuning Safety by Filtering Spurious Correlation

M. M. Salles*, **P. Goyal*** et al.
NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle. Poster presentation.

* Equal contribution.

INDUSTRY EXPERIENCE

Software Developer Aug 2022 – Dec 2022
Exacta Global Smart Solutions Philadelphia, PA

- Led a four-member team to build an ESP32 and AWS IoT Core traffic management system, improving efficiency by 40%; placed third among 50+ teams in the KETI oneM2M Hackathon.
- Automated AWS infrastructure using Terraform and CloudFormation, reducing setup time by 60% and achieving 99.5% uptime with a fault-tolerant design.

Machine Learning Engineer Jun 2021 – Aug 2021
Snapdeal Remote

- Built a Python and Scikit-learn forecasting model for server I/O operations, improving prediction accuracy by 40% to support resource allocation.
- Integrated predictive models with Linux server operations, reducing response times by 20%.

TEACHING EXPERIENCE

Graduate Teaching Assistant Aug 2022 – May 2024
Pennsylvania State University University Park, PA

- Led discussion sections and tutoring in discrete mathematics, covering logic, set theory, and graph theory; supported instruction and classroom management for 1,000 students.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, JavaScript, HTML/CSS
ML and data: TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib
Tools and platforms: Git, Linux, Docker, AWS IoT Core, Terraform, CloudFormation
Web and databases: Flask, React, Node.js, Express, MongoDB